Fading LED

```
int led = 9;
int brightness = 0;
int fadeAmount = 1;

void setup() {
  pinMode(led, OUTPUT);
}

void loop() {
  for (int i = 0; i < 200; i++) {
    digitalWrite(led, HIGH);
    delayMicroseconds(brightness + 1);
    digitalWrite(led, LOW);
    delayMicroseconds(256 - brightness);
  }
  brightness += fadeAmount;
 if (brightness >= 255 || brightness <= 0) {
    fadeAmount = -fadeAmount;
  }
}
```